PHYC10009 Physics Foundations

Electricity and Magnetism
Week 7 Lecture 3

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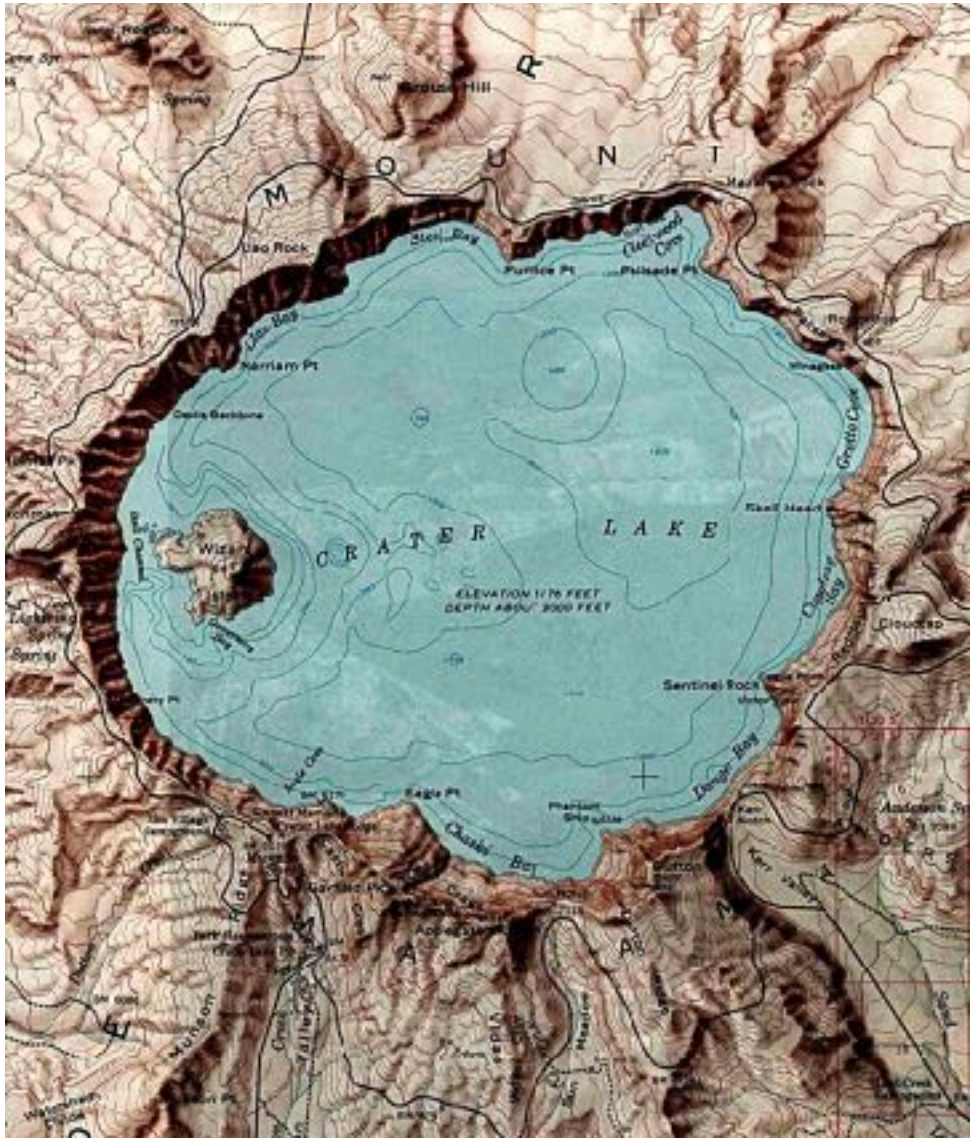
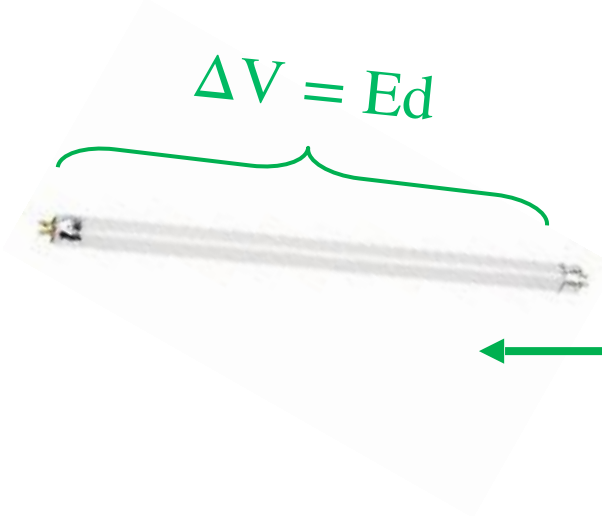
Electric potential V

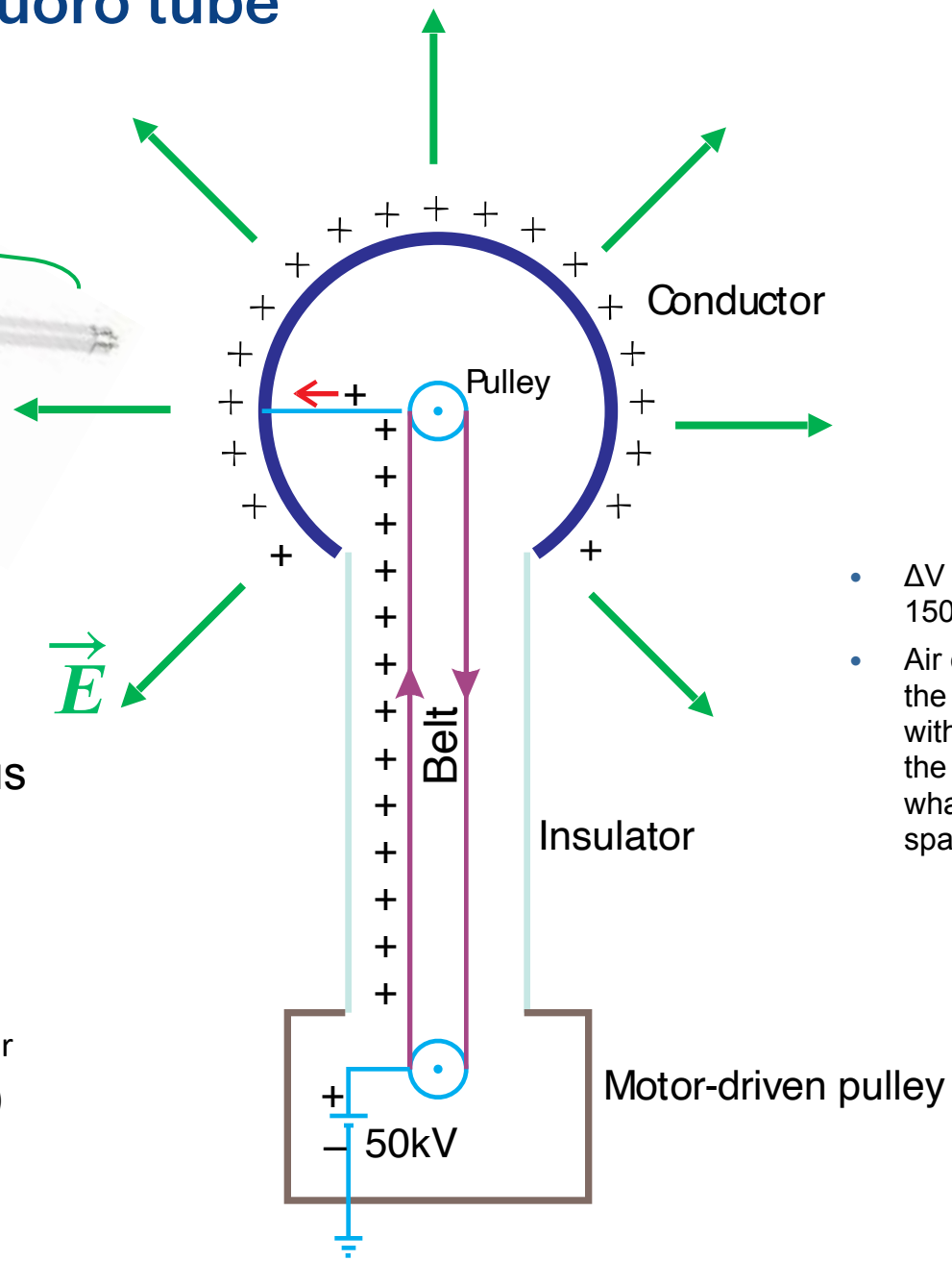
- Electric potential (voltage) is like height
- Lines of constant height are lines of constant gravitational potential



Van de Graaff + fluoro tube



$$\Delta V = Ed$$



- ΔV can be 50,000 to 150,000 V!
- Air emits light when the ions recombine with electrons from the wand. This is what we see when a spark occurs.

Note: E depends on radius

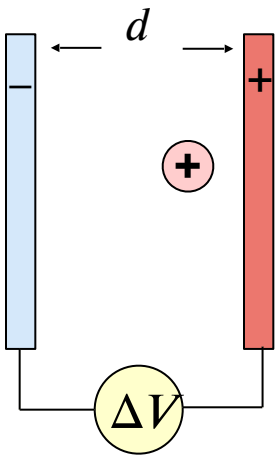
so $\Delta V = \int E(r)dr$

D El-8 Van de Graaff Generator (Lighting a Fluorescent Tube)

17.1, 17.2 Electric potential V and electric fields E

The electric field strength is given by the *gradient* of electric potential

We have $\Delta V = -Ed$ so $E = -\dfrac{\Delta V}{d}$



* Warning: only component of d along field lines (up and down the hill) , not perpendicular component

Electric field has units $[E] = \dfrac{[\Delta V]}{[d]} = \mathrm{V/m}$ (volts per metre)

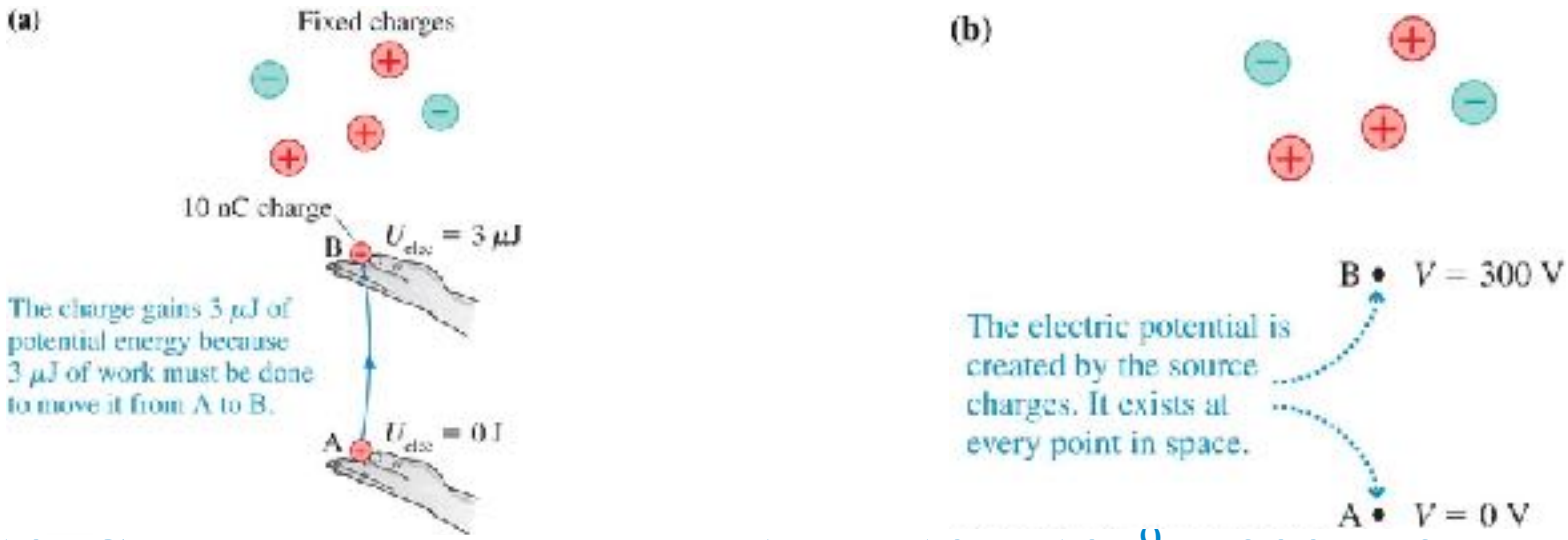
Use electric potential to calculate

- Change in energy of a charge
- Size of electric field
- Strength of electric current

Electric potential (voltage) V

- Electric potential is a *scalar field* defined at all positions in space
  - like temperature in a house: different in different places, no direction
- Given electric potential, the potential energy of a charge is defined

- Fig (a) shows work done in moving 10nC charge from a to B.
  How much work is required to move a 1μC charge from A to B?
  Hard to calculate using Coulomb's law but easy if we know V at A and B:



For 10nC we have: $W = qEd = q\Delta V = 10 \times 10^{-9} \times 300 = 3\mu J$

For $1\ \mu C$ we have: $W = qEd = q\Delta V = 1 \times 10^{-6} \times 300 = 300\ \mu J$

17.4 Example: the electron volt (unit of energy)

The charge on an electron is $-1.6 \times 10^{-19}\mathrm{C}$

If the electron is accelerated though a potential difference ΔV of 1 volt, what is the energy gain of the electron?

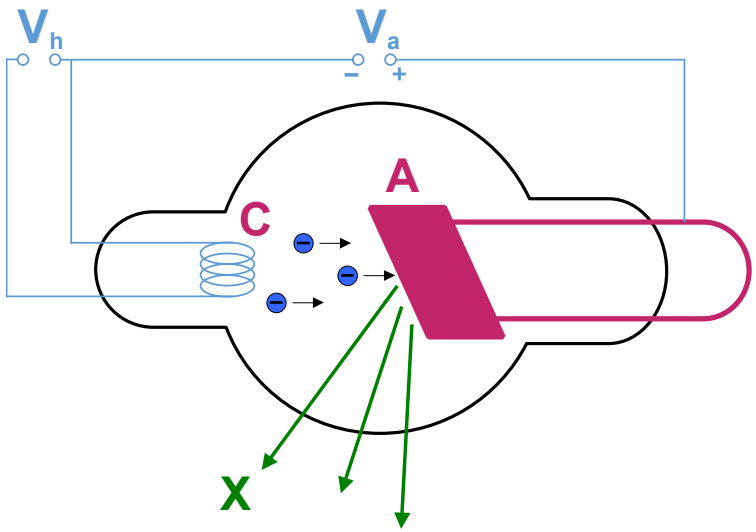
Energy $\mathrm{PE} = q\Delta V = 1.6 \times 10^{-19} \times 1 = 1.6 \times 10^{-19}\ \mathrm{J}$

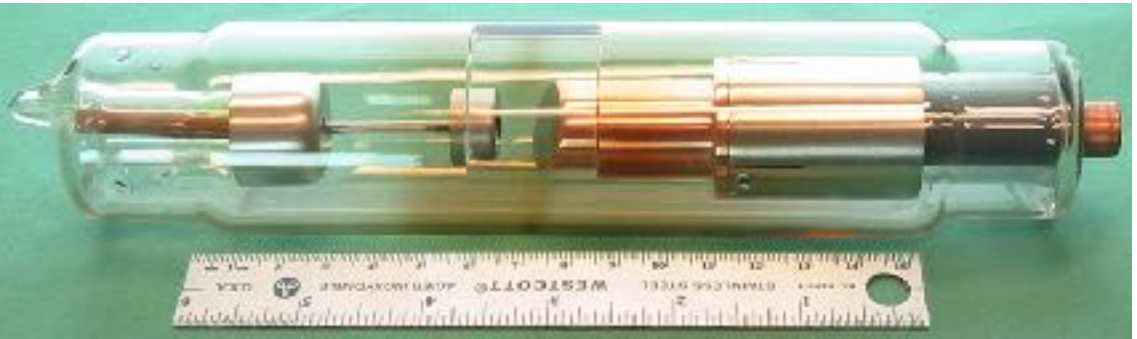
The electron-Volt (eV): a unit of <u>energy</u> (***not*** potential)

$1\mathrm{eV}$	$1.6 \times 10^{-19}\mathrm{J}$	Approx energy of visible photon
$1\mathrm{keV}$	$10^{3}\mathrm{eV}$	Low energy x-rays
$30\mathrm{keV}$	$3 \times 10^{4}\mathrm{eV}$	Old-fashioned colour TV
$100\mathrm{keV}$	$10^{5}\mathrm{eV}$	Medical x-rays (25 to 140keV)
$1\mathrm{MeV}$	$10^{6}\mathrm{eV}$	Nuclear processes, radiation
$1\mathrm{GeV}$ – 1TeV	$10^{9} - 10^{12}\mathrm{eV}$	Cosmic rays, particle physics
	$10^{21}\mathrm{eV}$	Cosmic rays from AGN

Example: energy PE and electric potential V

- In a medical x-ray tube, electrons are accelerated at up to 100,000 volts and then collide with a tungsten anode
- What is the final velocity of the electrons?
- What is the maximum energy of the x-rays?





KE $\frac{1}{2}mv^2 = \text{PE} = qV_a$

$$v^2 = \frac{2qV_a}{m}$$

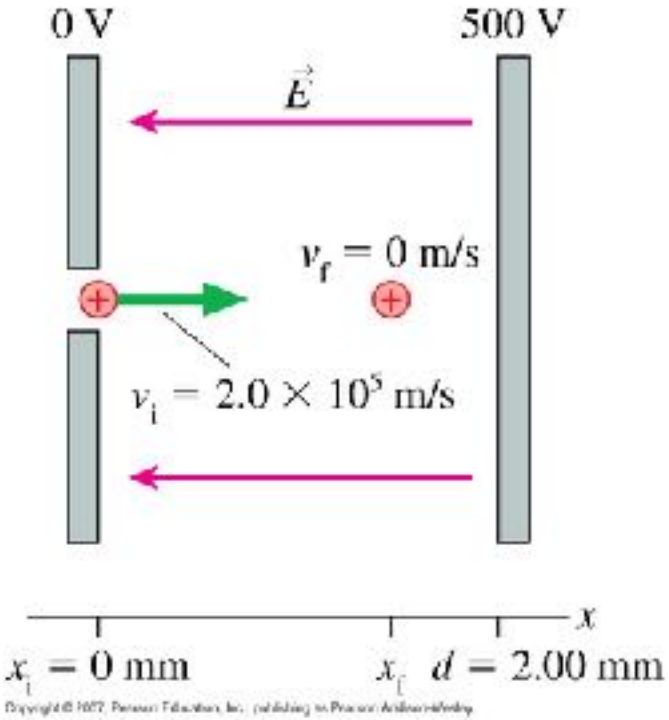
$$v = \sqrt{\frac{2qV_a}{m}}$$

$$= 1.9 \times 10^8 \text{ m/s} = 62\% \ c$$

Speed of light $c = 3 \times 10^8 \text{m/s}$

Example: using electric potential V

- A proton, velocity 2×10^5 m/s enters a uniform electric field as shown
  - Will the proton reach the anode (+500V)?
  - If not, how far does it travel before returning?



Energy of proton :

$$\frac{1}{2}mv^2 = 0.5 \times 1.67 \times 10^{-27} \times 4 \times 10^{10}$$

$$= 3.34 \times 10^{-17}\text{J} \quad \text{and divide by } q_e \text{ to find}\ldots$$

$$= 210\text{eV}$$

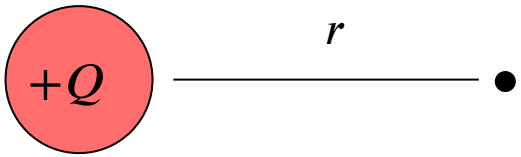
Thus proton does ***not*** have enough energy.

Will turn around at 210V i.e.

$$x = \left(\frac{210}{500}\right) \times 2.0\text{mm} = 0.84\text{mm}$$

17.5 Example: electric potential around a point charge

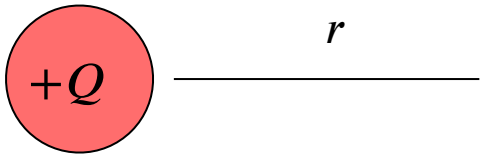
Push "test charge" q' towards fixed charge q until radius R



$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r} = k\frac{Q}{r}$$ for a point charge Q

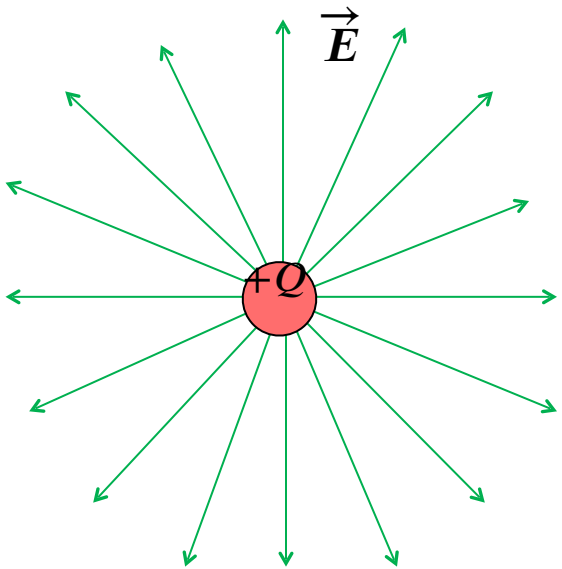
Electric potential and electric field

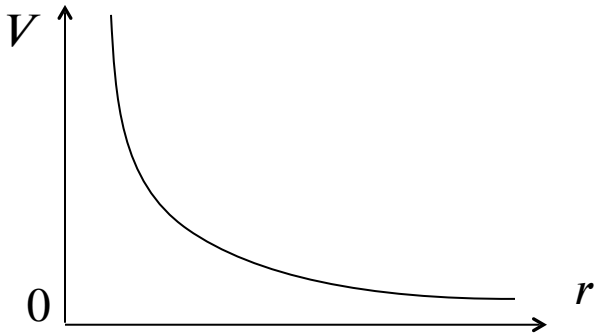
- Potential is + near + charges



$$V = \frac{1}{4\pi\epsilon_0}\frac{Q}{r}$$

- Potential *decreases* when moving in direction of electric field

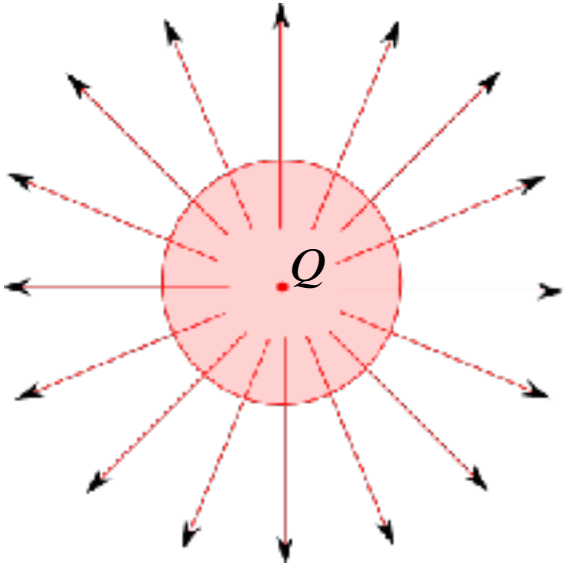




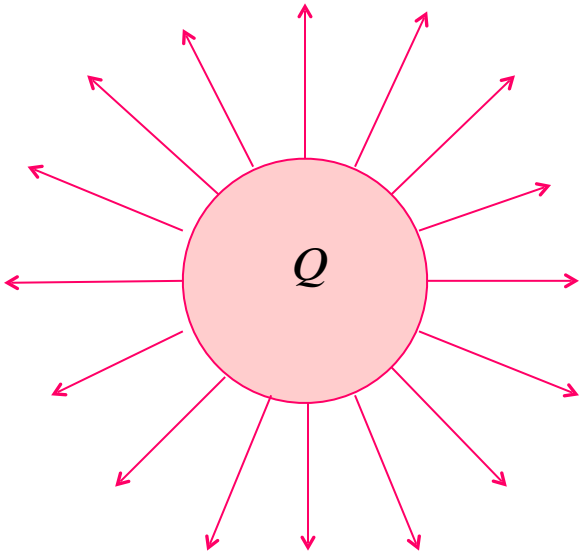
https://phet.colorado.edu/sims/html/charges-and-fields/latest/charges-and-fields_en.html

Example: electric potential of a sphere

Spherical symmetry: electric field <u>outside</u> a charged sphere is identical to electric field from a point charge



$$V = k\frac{Q}{r}$$



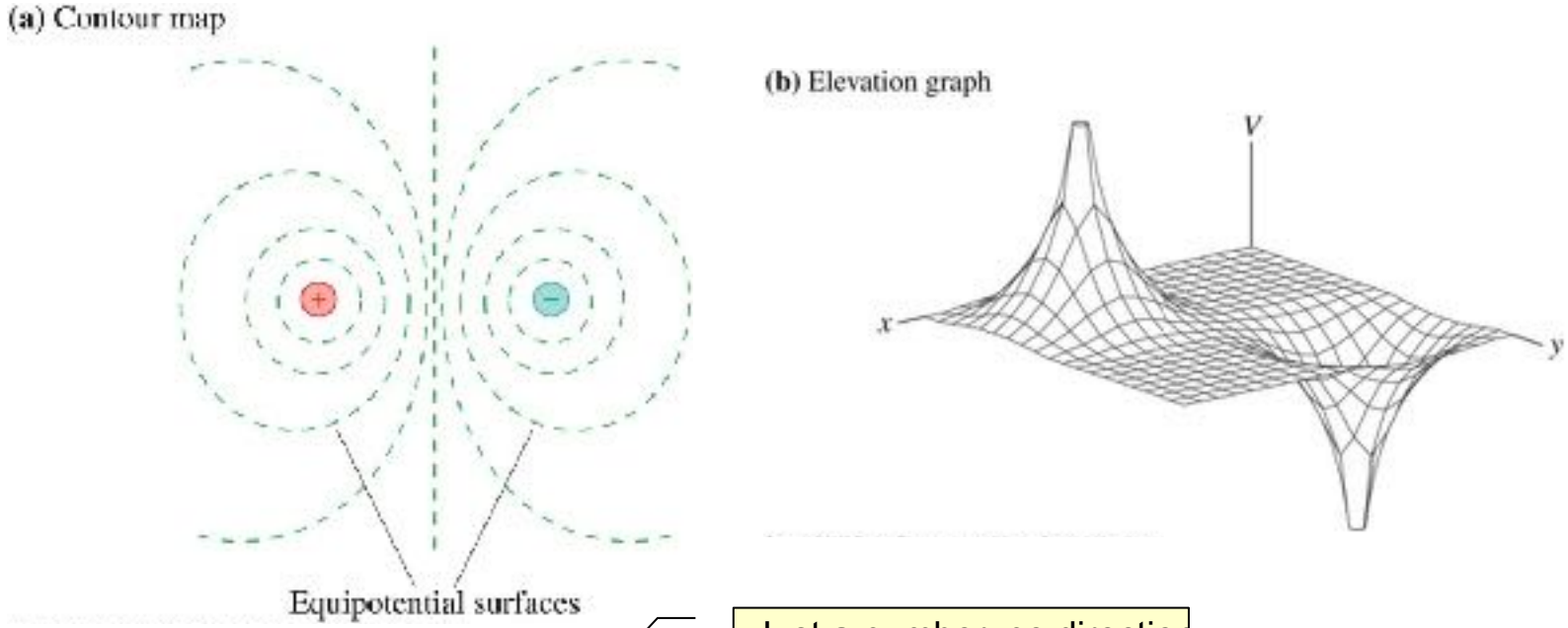
Electric *potential* outside a uniformly charged sphere is identical to that of a point charge

Superposition: electric potential of many charges

The potential for a collection of point charges is just the sum of the potential from each

$$V = \sum_i k \frac{Q_i}{r_i}$$

where r_i is the distance from charge Q_i to the point of interest



Just a number, no direction

Note scalar sum: potential is a scalar quantity (electric field is a vector quantity)

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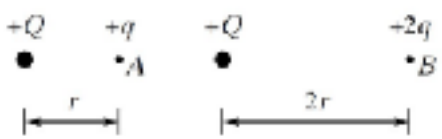
Electric Potential

Two test charges are brought separately near a charge +Q.
First, test charge $+q$ is brought to point A at distance r from +Q.
Next, $+q$ is removed and a test charge $+2q$ is brought to point B at distance $2r$ from +Q.



Compared with the electric potential of the charge at A, the electric potential of the charge at B is

greater

smaller

the same

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Potential in conductor

A solid spherical conductor is given a net nonzero charge. The electric potential of the conductor is

- largest at the centre
- largest at the surface
- Largest somewhere between centre and surface
- constant throughout the volume

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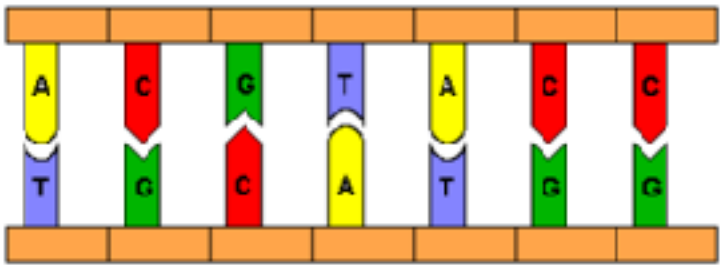
Exam 2015

Question 7

25 marks

(a) The strands of a DNA molecule are bound together by electrostatic attraction. Thymine T on one strand binds to adenine A on the other, and similarly cytosine C and guanine G groups are attracted to each other (see figure).



The TA bond can be characterised as charges of $-0.4e$ and $+0.4e$ on the T and A groups, separated by distance 0.18 nm, where $e = 1.60 \times 10^{-19}$C. Similarly, the CG bond is formed by charges of $\pm 0.2e$ separated by 0.20 nm.

(i) Calculate the force of each TA and each CG bond.

(ii) There are of order 10^5 TA and 10^5 GC bonds in a DNA molecule. Calculate the total potential energy of the strand-binding in a DNA molecule.

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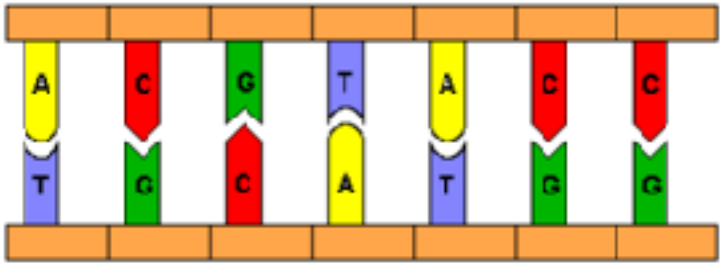

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(ii) There are of order 10^5 TA and 10^5 GC bonds in a DNA molecule. Calculate the total potential energy of the strand-binding in a DNA molecule.

(i)

$$F_{TA} = -\frac{kQ_1Q_2}{r^2} = \frac{9 \times 10^9 \times (0.4 \times 1.6 \times 10^{-19})^2}{(0.18 \times 10^{-9})^2} = 1.14 \times 10^{-9} N$$

$$F_{CG} = -\frac{kQ_1Q_2}{r^2} = \frac{9 \times 10^9 \times (0.20 \times 1.6 \times 10^{-19})^2}{(0.2 \times 10^{-9})^2} = 2.30 \times 10^{-10} N$$

(ii)

$$PE_{TA} = -\frac{qQ_1Q_2}{r} = -\frac{9 \times 10^9 \times (0.4 \times 1.6 \times 10^{-19})^2}{0.18 \times 10^{-9}} = -2.05 \times 10^{-19} J$$

$$PE_{CG} = -\frac{qQ_1Q_2}{r} = -\frac{9 \times 10^9 \times (0.2 \times 1.6 \times 10^{-19})^2}{0.20 \times 10^{-9}} = -4.61 \times 10^{-20} J$$

$$PE = 10^5 \times (-2.05 \times 10^{-19} - 4.61 \times 10^{-20}) = -2.51 \times 10^{-14} J = -1.56 \times 10^5 eV$$

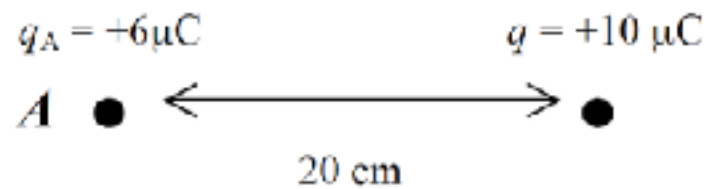
Example: 2008 exam question

voltage

(b) Consider the electric potential and electric potential energy of the following situations.

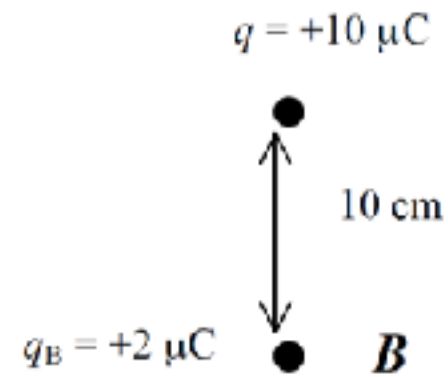
Scenario A:

A charge $q_A = +6\ \mu C$ is placed at position A, 20cm from a charge $q = +10\ \mu C$ as shown in the figure at right.



Scenario B:

The charge q_A is removed and a charge $q_B = +2\ \mu C$ is placed at position B, 10 cm from the $+10\ \mu C$ charge (see diagram).



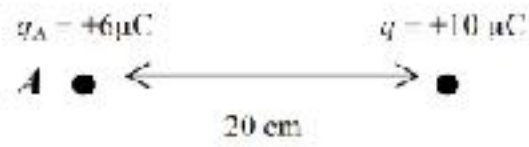
voltage

(i) At which location (A or B) is the electric potential greater? (Justify your answer)

(ii) Which charge (q_A or q_B) has the higher electrostatic potential energy? (Justify your answer)

2008 Exam answers

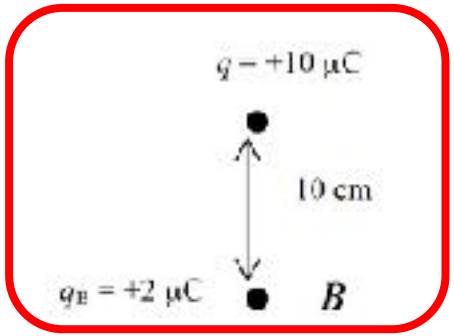
$$\mathrm{PE}(r) = k\frac{Qq}{r} \qquad V = \frac{\mathrm{PE}}{q}$$



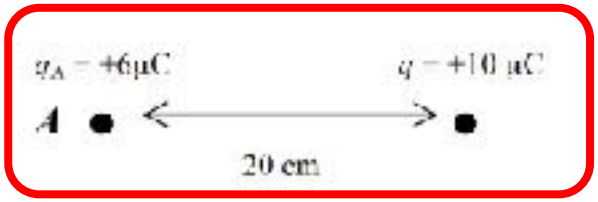
(i)

$$V_A = \frac{kQ_2}{r} = \frac{k \times 10 \times 10^{-6}}{0.2} = k \times 50 \times 10^{-6} V$$

$$V_A = \frac{kQ_2}{r} = \frac{k \times 10 \times 10^{-6}}{0.1} = k \times 100 \times 10^{-6} V$$

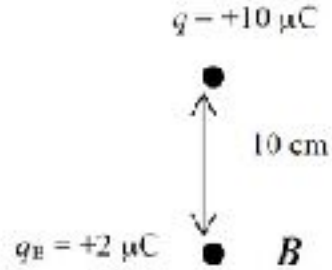


(ii)



$$PE_A = \frac{kq_AQ_2}{r} = \frac{k \times 6 \times 10^{-6} \times 10 \times 10^{-6}}{0.2} = 300 \times 10^{-12} J$$

$$PE_A = \frac{kq_BQ_2}{r} = \frac{k \times 2 \times 10^{-6} \times 10 \times 10^{-6}}{0.1} = 200 \times 10^{-12} J$$



Example: 2009 exam

Question 4 **[2 + 2+ 2 + 3 + 3 + 4 = 16 marks]**

(a) A proton is held very close to the surface of a 1.0 cm diameter sphere charged so that its surface is at +1000V

(i) What is the total charge on the sphere?

(ii) What is the magnitude of the electric field at the surface of the sphere?

(iii) What is the force on the proton?

(iv) The proton is released and is accelerated away. What is the final velocity of the proton?

Example: 2009 exam

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(iii) What is the force on the proton?

(iv) The proton is released and is accelerated away. What is the final velocity of the proton?

(i) Electric potential of a sphere is the same as point change

$$V = \frac{kQ}{r} = \frac{9 \times 10^9 \times Q}{0.005} = 10^3 V \Rightarrow Q = \frac{0.005 \times 10^3}{9 \times 10^9} = 5.55 \times 10^{-10} C$$

(ii)

$$E = \frac{kQ}{r^2} = 2 \times 10^5 \ V/m$$

(iii)

$$F = qE = 1.6 \times 10^{-19} \times 2 \times 10^5 = 3.2 \times 10^{-14} N$$

(iv) Convert PE of proton near sphere to KE of motion, so KE=PE

$$PE = qV = 1.6 \times 10^{-19} \times 1 \times 10^3 = 1.6 \times 10^{-16} J$$

$$KE = \frac{1}{2} m v^2 = 1.6 \times 10^{-16} \Rightarrow v^2 = \frac{2 \times 1.6 \times 10^{-16}}{1.67 \times 10^{-27}} = 1.93 \times 10^{11} \Rightarrow v = 4.4 \times 10^5 \ ms^{-1}$$

GAMSAT Questions

Three small identical metal spheres carry electric charges, +Q, +2Q, and +3Q. The centres of the spheres lie in a straight line at fixed positions R, S and T, such that the distance between the spheres at R and S is d_1 and that between the spheres at S and T is d_2 (Figure 1). Assume that the distances between the spheres is large in comparison with their size, and that the charges on the spheres act like point charges.

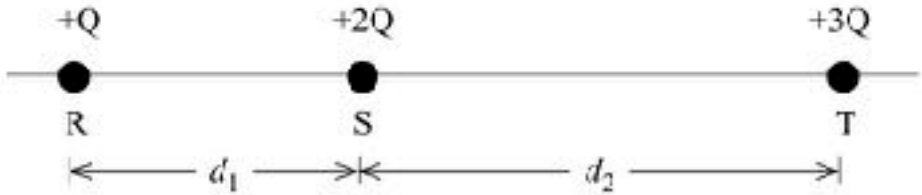

Figure 1

Q1. The sphere at S will experience a zero net electric force when d_2 is equal to:

A. $3d_1$ **B.** $\frac{d_1}{3}$ **C.** $\sqrt{3}d_1$ **D.** $\frac{d_1}{\sqrt{3}}$

Q2. A small metal sphere (identical to those at R,S,T) carrying a charge of -2Q is first brought into contact with the sphere at R, and then into contact with the sphere at S, and finally into contact with the sphere at T.

After making contact with the three spheres, in the stated order, the charge on the sphere at T is

A. $+3Q$ **B.** $+\frac{15Q}{4}$ **C.** $+\frac{15Q}{8}$ **D.** $+4Q$

GAMSAT answered..

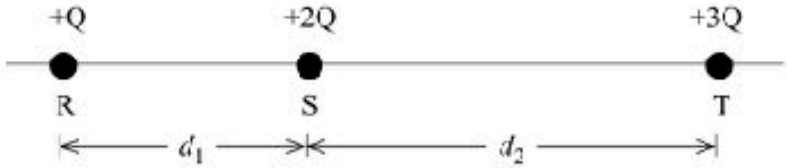

Figure 1

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A. $3d_1$ **B.** $\frac{d_1}{3}$ **C.** $\sqrt{3}d_1$ **D.** $\frac{d_1}{\sqrt{3}}$

$$F_{RS} = F_{ST} \Rightarrow \frac{k2Q^2}{d_1^2} = \frac{k6Q^2}{d_2^2} \Rightarrow d_2^2 = \frac{6d_1^2}{2} = 3d_1^2 \Rightarrow d_2 = \sqrt{3}d_1$$

Q2. A small metal sphere (identical to those at R,S,T) carrying a charge of -2Q is first brought into contact with the sphere at R, and then into contact with the sphere at S, and finally into contact with the sphere at T.
After making contact with the three spheres, in the stated order, the charge on the sphere at T is

A. $+3Q$ **B.** $+\frac{15Q}{4}$ **C.** $+\frac{15Q}{8}$ **D.** $+4Q$

Identical spheres => charged shared equally to both.

$$-2Q + Q = -Q \Rightarrow \frac{-Q}{2} \text{after touching R}$$

$$-Q/2 + 2Q = 3Q/2 \Rightarrow \frac{3Q}{4} \text{after touching S}$$

$$3Q/4 + 3Q = 15Q/4 \Rightarrow \frac{15Q}{8} \text{on T}$$

Summary: electric potential

- Electric potential energy $\Delta \mathrm{PE} = Fd = qEd$ (only if *E* is uniform; careful of sign)
- Electric potential energy around point charge Q $\mathrm{PE}(r) = k\dfrac{Qq}{r}$
- Electric potential = potential energy per unit charge $V = \dfrac{\mathrm{PE}}{q}$
- Electric field from potential $V = Ed$
- Potential around a point charge $V = k\dfrac{Q}{r}$
- The electron volt unit of energy $1\mathrm{eV} = 1.6 \times 10^{-19}\mathrm{J}$